

Model appendix

This appendix contains supplemental figures and detailed model tables for the multiscale spectral–biodiversity analysis.

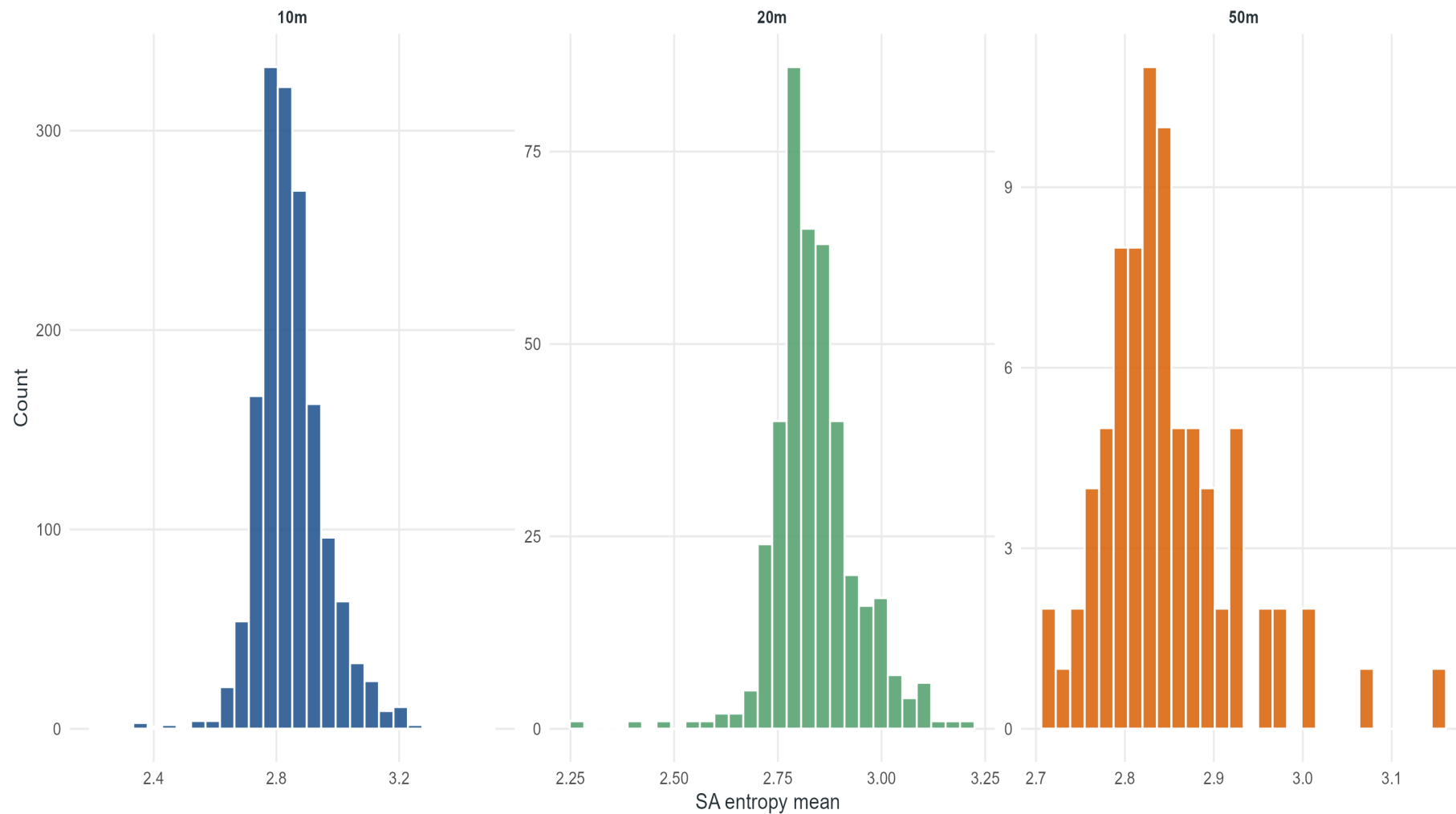
All models use standardized responses and predictors. Candidate sets include null, single biodiversity, environment–only, biodiversity plus environment, a

Tables are written in full to `reports/tables/multiscale\_spectral\_biodiversity/`.

SA entropy mean distribution

SA entropy mean distribution by scale

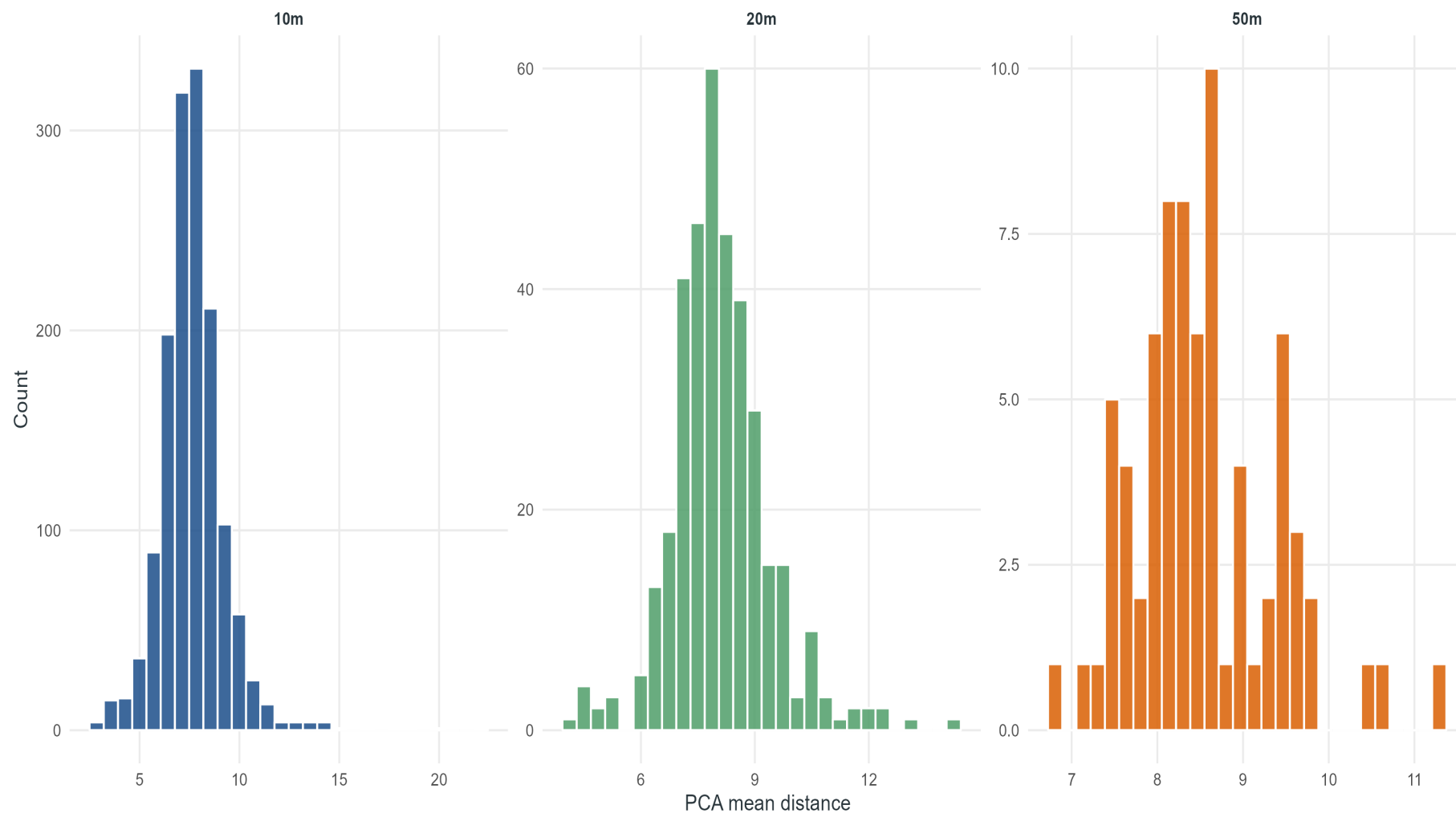
Most quadrat values are means of 70 bootstrap iterations using up to 5,000 retained pixels; the small exact subset used all retained pixel pairs.



PCA mean distance distribution

PCA mean distance distribution by scale

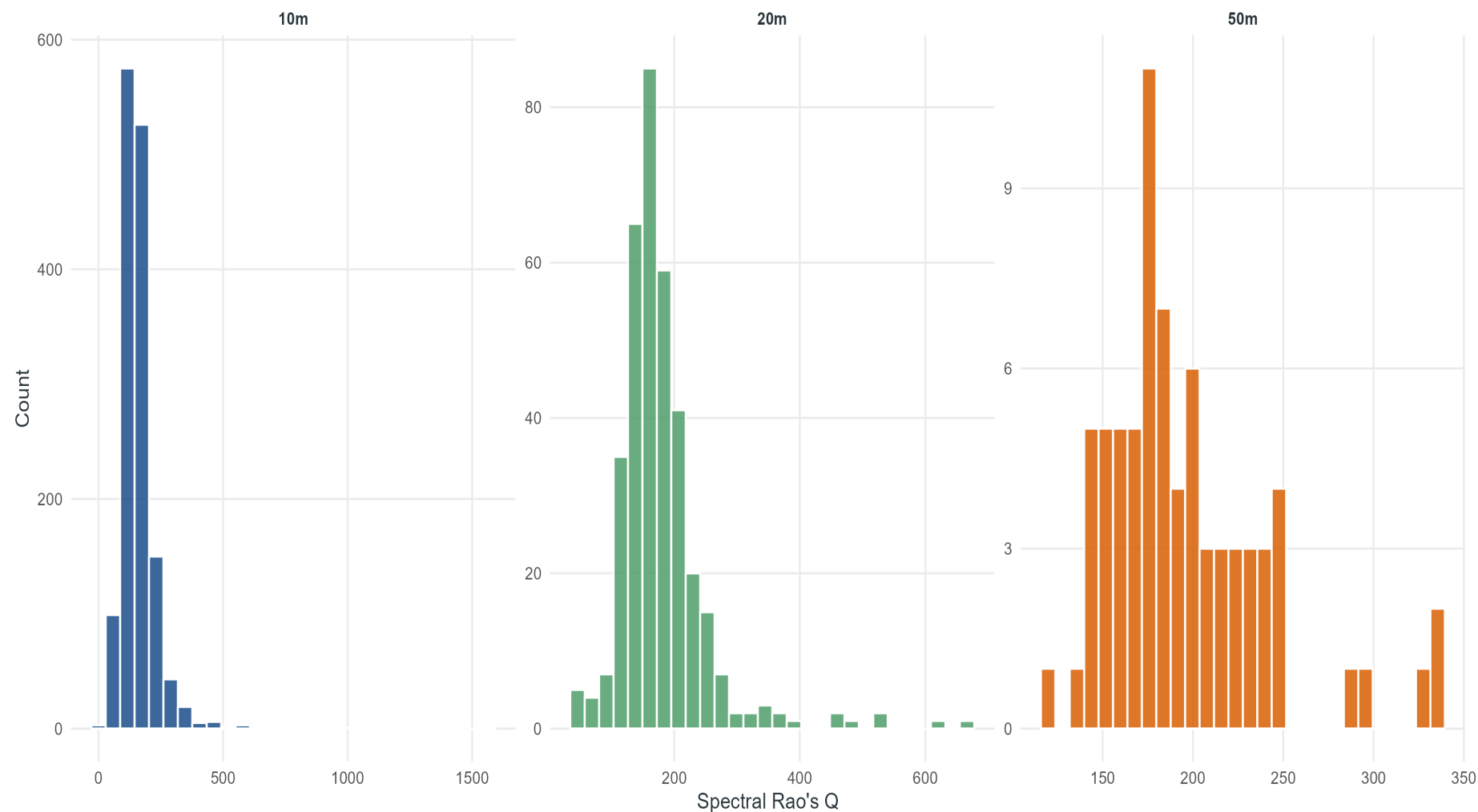
Distributions are shown for primary-analysis quadrats with non-missing values.



Spectral Rao's Q distribution

Spectral Rao's Q distribution by scale

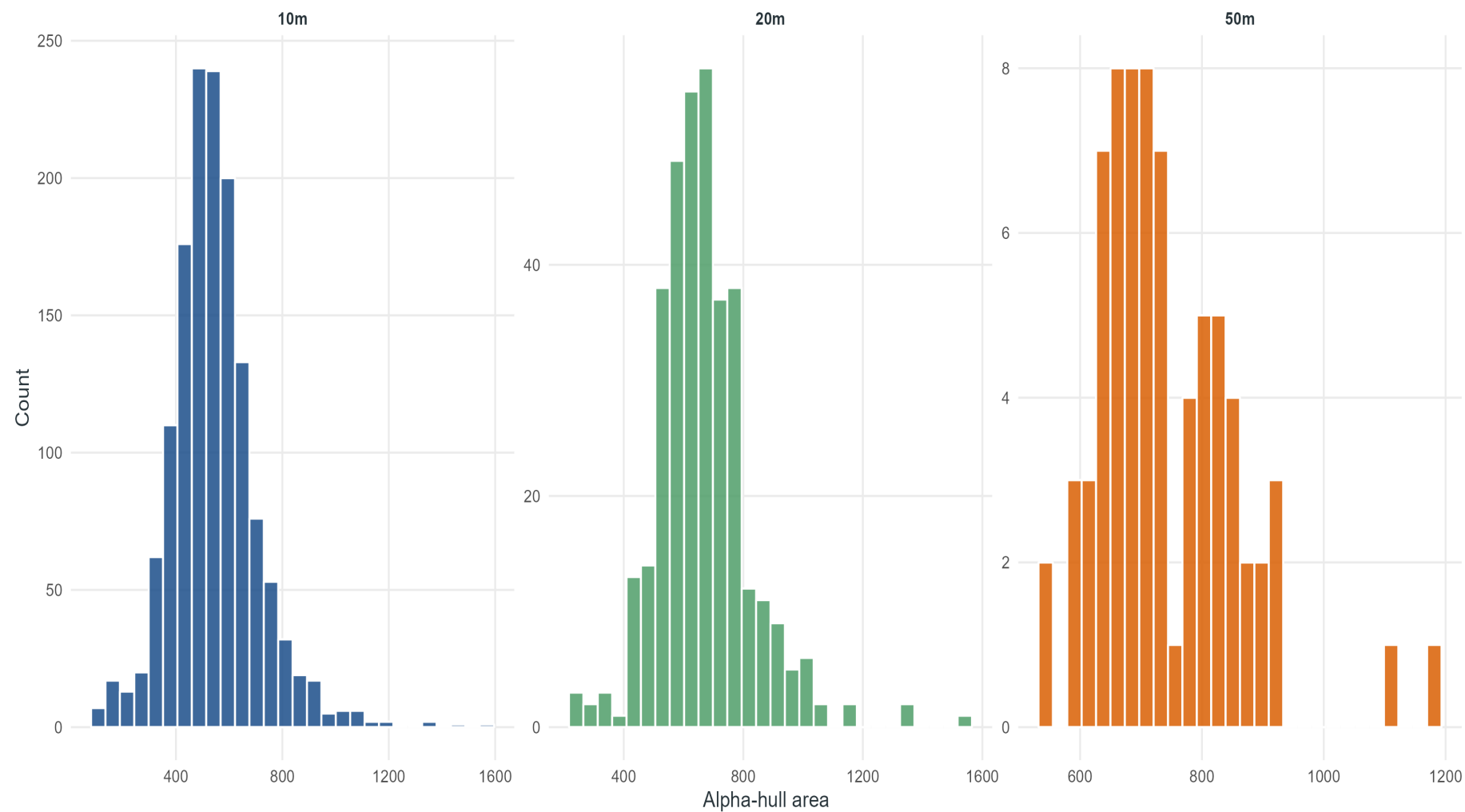
Distributions are shown for primary-analysis quadrats with non-missing values.



Alpha-hull area distribution

Alpha-hull area distribution by scale

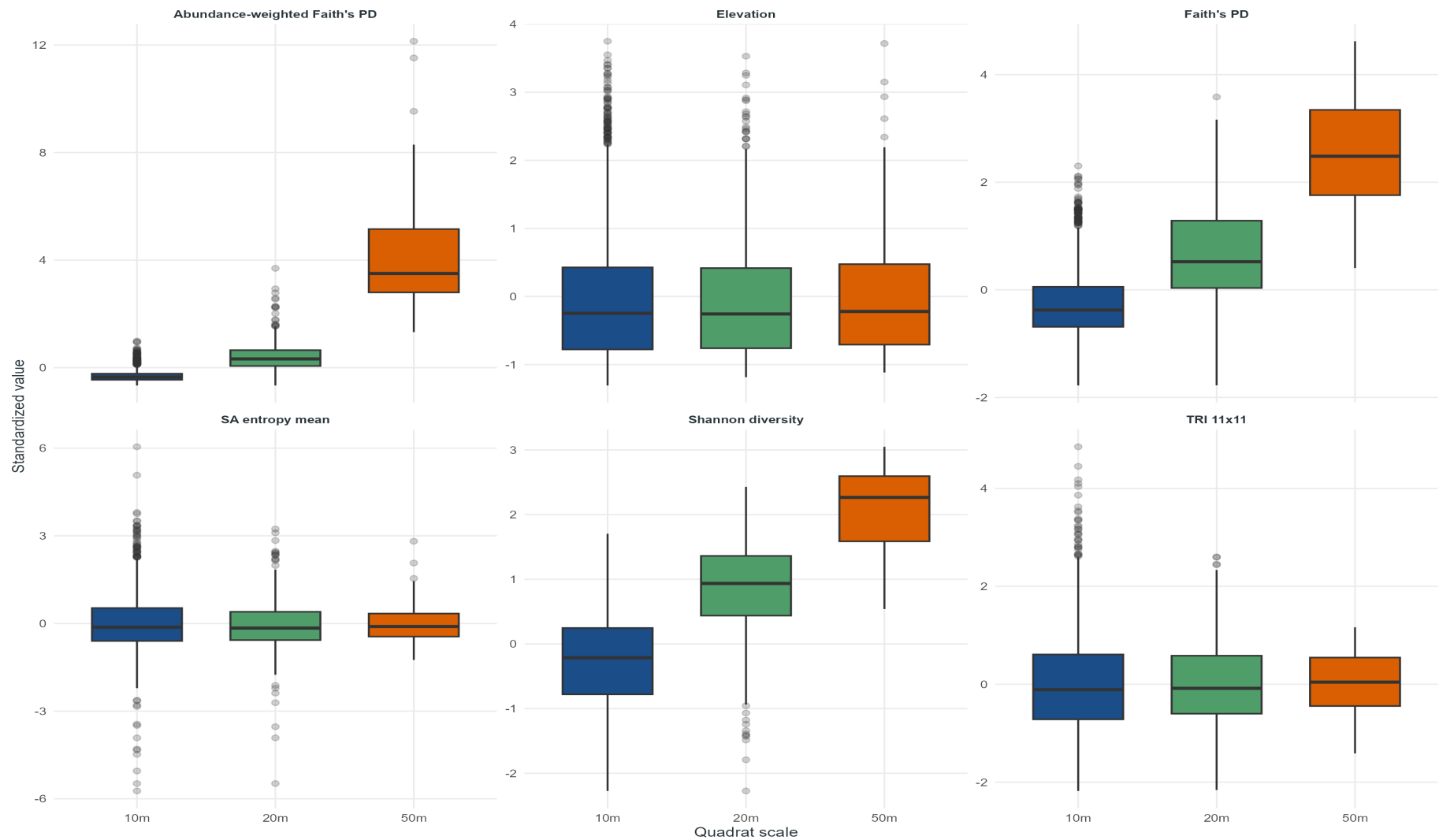
Distributions are shown for primary-analysis quadrats with non-missing values.



Scale comparison of standardized variables

Scale comparison of standardized variables

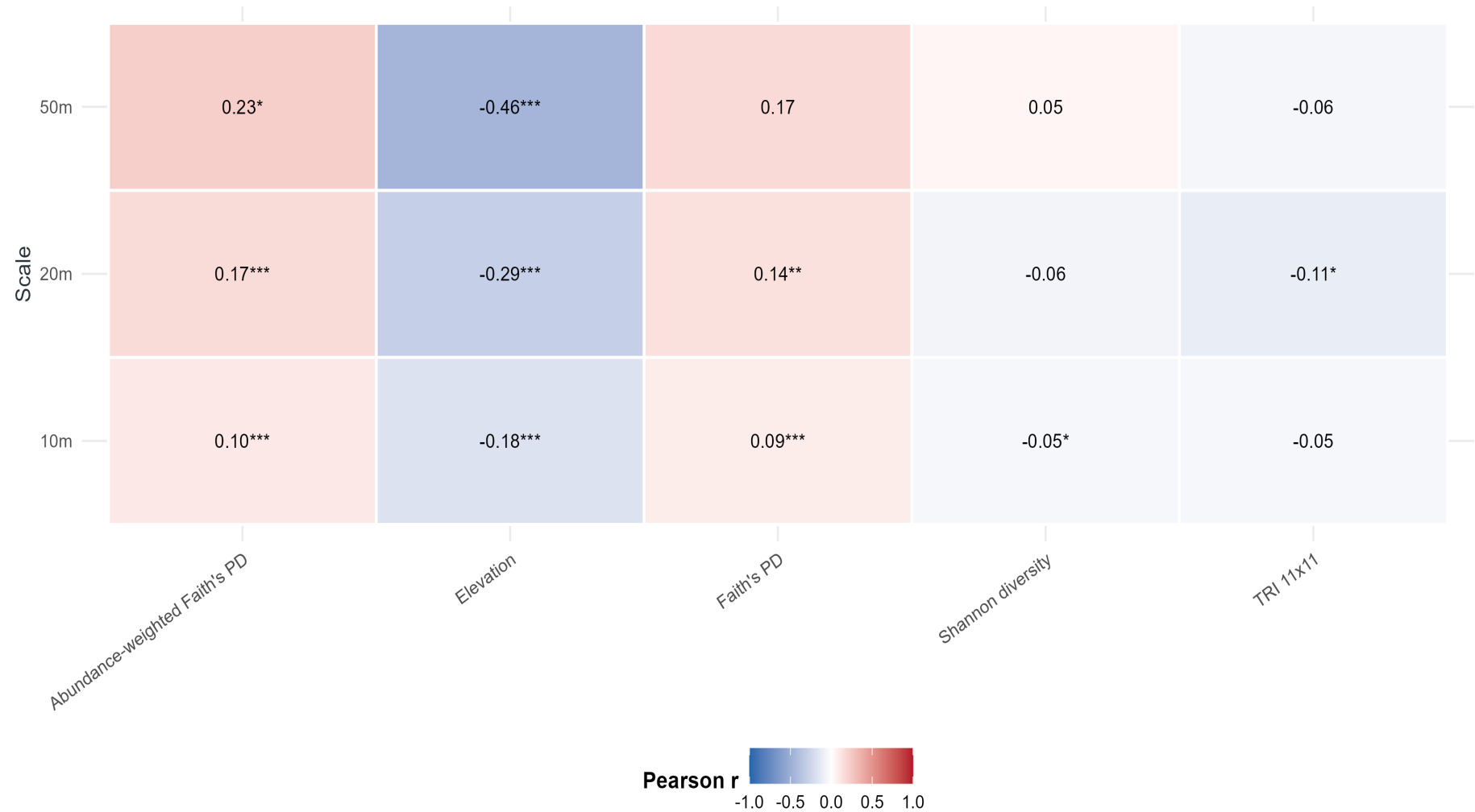
Values are standardized within each variable to make scale shifts visually comparable.



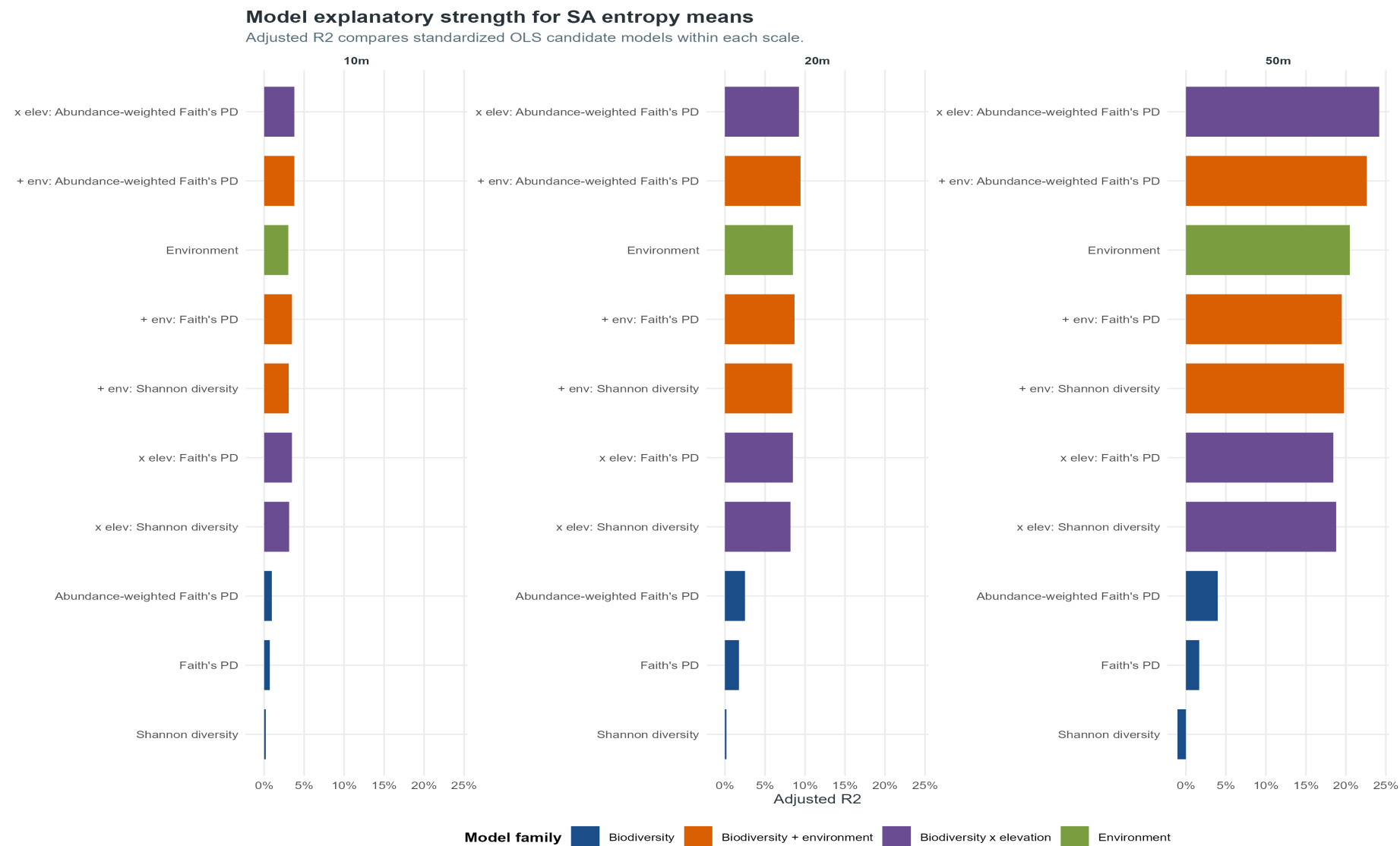
Primary correlations

Primary correlations with SA entropy means

Pearson r values for primary-analysis quadrats. Asterisks mark $p < 0.05$, 0.01 , and 0.001 .



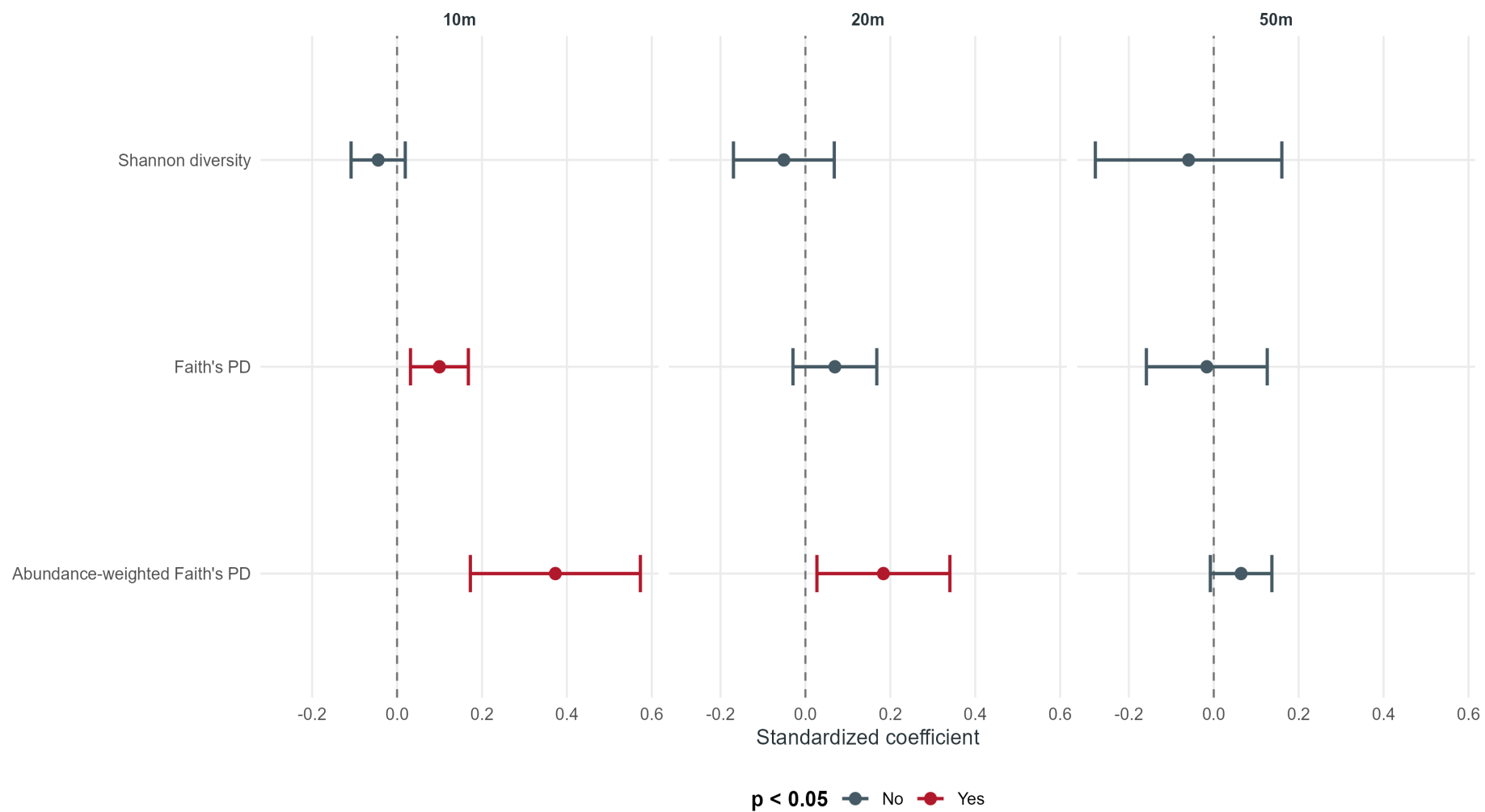
Adjusted R2 comparison



Coefficient forest

Standardized biodiversity effects after environmental controls

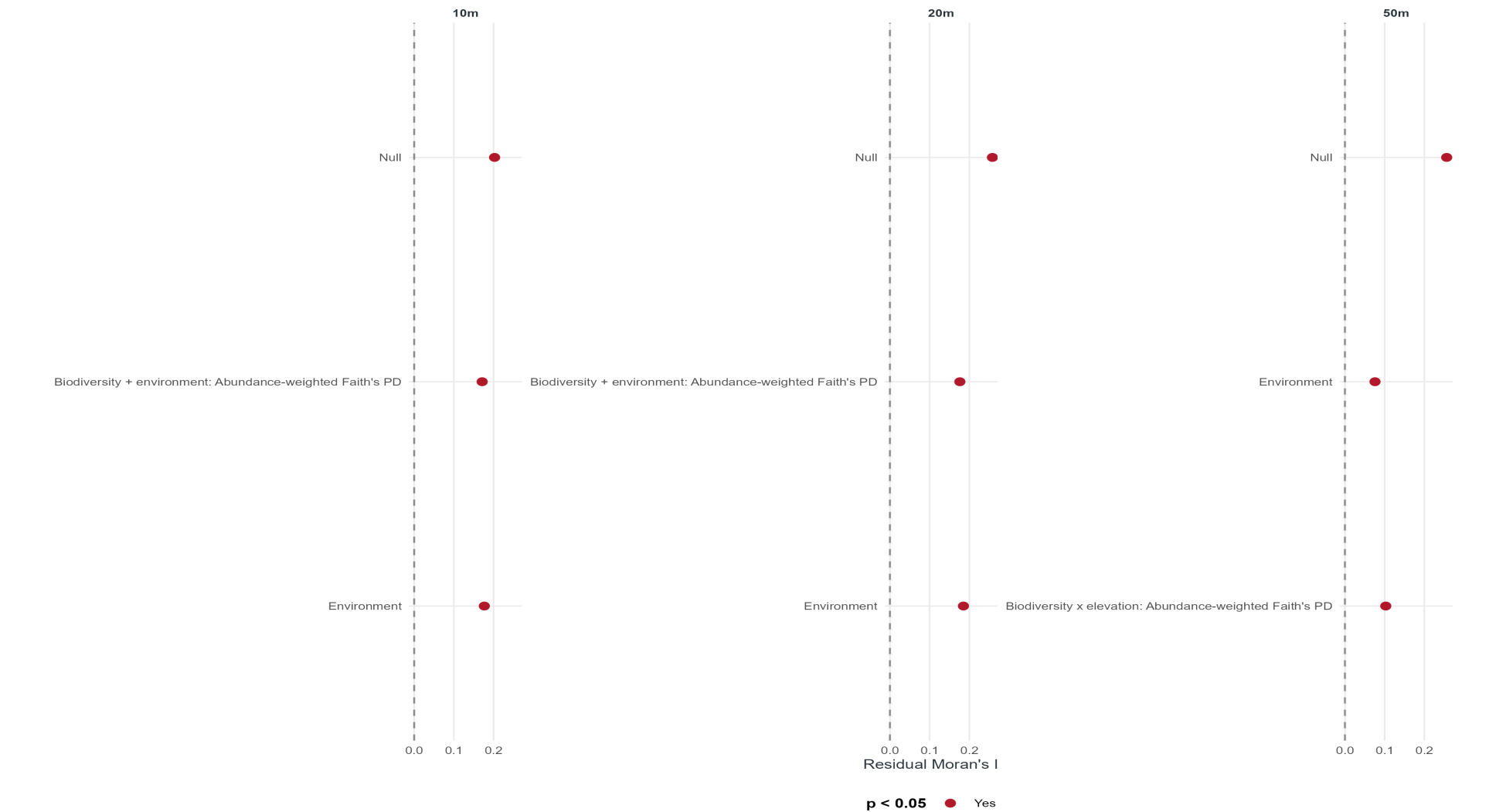
Points are standardized coefficients; bars are 95 percent confidence intervals.



Residual Moran's I diagnostics

Spatial autocorrelation in model residuals

Moran's I is calculated from 8-nearest-neighbor residuals. Significant residual structure cautions against over-interpreting p-values.



Top candidate models by response and scale

Scale	Response	Model	n	Adj. R2	AIC	Delta AIC
10m	Alpha-hull area	Biodiversity + environment: Abundance-weighted Faith's PD	1439	0.004	4047.3	0.0
10m	Alpha-hull area	Environment	1439	0.003	4047.7	0.4
10m	Alpha-hull area	Biodiversity + environment: Shannon diversity	1439	0.004	4047.9	0.6
10m	Alpha-hull area	Biodiversity: Abundance-weighted Faith's PD	1439	0.002	4048.7	1.4
20m	Alpha-hull area	Biodiversity: Abundance-weighted Faith's PD	360	0.041	981.8	0.0
20m	Alpha-hull area	Biodiversity + environment: Abundance-weighted Faith's PD	360	0.046	982.0	0.3
20m	Alpha-hull area	Biodiversity + environment: Shannon diversity	360	0.026	989.4	7.7
20m	Alpha-hull area	Biodiversity + environment: Faith's PD	360	0.022	990.8	9.0
50m	Alpha-hull area	Biodiversity: Abundance-weighted Faith's PD	74	0.160	149.8	0.0
50m	Alpha-hull area	Biodiversity + environment: Abundance-weighted Faith's PD	74	0.174	150.4	0.6
50m	Alpha-hull area	Biodiversity + environment: Shannon diversity	74	0.115	155.6	5.8
50m	Alpha-hull area	Environment	74	0.070	158.3	8.5
10m	PCA mean distance	Environment	1440	0.004	4307.3	0.0
10m	PCA mean distance	Biodiversity + environment: Abundance-weighted Faith's PD	1440	0.005	4307.6	0.3
10m	PCA mean distance	Biodiversity + environment: Shannon diversity	1440	0.005	4308.1	0.8
10m	PCA mean distance	Biodiversity + environment: Faith's PD	1440	0.004	4308.8	1.6
20m	PCA mean distance	Biodiversity: Abundance-weighted Faith's PD	360	0.022	912.9	0.0
20m	PCA mean distance	Biodiversity + environment: Abundance-weighted Faith's PD	360	0.023	914.3	1.5
20m	PCA mean distance	Biodiversity + environment: Shannon diversity	360	0.015	917.2	4.3
20m	PCA mean distance	Biodiversity: Shannon diversity	360	0.008	917.8	4.9
50m	PCA mean distance	Biodiversity + environment: Abundance-weighted Faith's PD	74	0.155	113.8	0.0
50m	PCA mean distance	Biodiversity: Abundance-weighted Faith's PD	74	0.129	114.1	0.3
50m	PCA mean distance	Biodiversity + environment: Shannon diversity	74	0.073	120.6	6.8
50m	PCA mean distance	Environment	74	0.029	123.1	9.3
10m	Spectral Rao's Q	Biodiversity + environment: Shannon diversity	1440	0.005	4264.3	0.0
10m	Spectral Rao's Q	Biodiversity: Shannon diversity	1440	0.003	4265.3	1.0

Primary-response coefficients

Scale	Model	Term	Estimate	SE	CI low	CI high	p
10m	Biodiversity: Abundance-weighted Faith's PD	phy_afaith	0.420	0.103	0.219	0.621	<0.001
10m	Biodiversity: Faith's PD	phy_faith	0.123	0.035	0.054	0.192	<0.001
10m	Biodiversity: Shannon diversity	sp_shannon	-0.066	0.033	-0.130	-0.002	0.043
10m	Biodiversity + environment: Abundance-weighted Faith's PD	env_elev	-0.199	0.029	-0.256	-0.141	<0.001
10m	Biodiversity + environment: Abundance-weighted Faith's PD	env_tri11	-0.003	0.025	-0.052	0.046	0.910
10m	Biodiversity + environment: Abundance-weighted Faith's PD	phy_afaith	0.373	0.102	0.173	0.573	<0.001
10m	Biodiversity + environment: Faith's PD	env_elev	-0.196	0.029	-0.253	-0.138	<0.001
10m	Biodiversity + environment: Faith's PD	env_tri11	-0.012	0.025	-0.060	0.037	0.643
10m	Biodiversity + environment: Faith's PD	phy_faith	0.100	0.035	0.032	0.168	0.004
10m	Biodiversity + environment: Shannon diversity	env_elev	-0.200	0.029	-0.258	-0.143	<0.001
10m	Biodiversity + environment: Shannon diversity	env_tri11	-0.010	0.025	-0.059	0.039	0.697
10m	Biodiversity + environment: Shannon diversity	sp_shannon	-0.044	0.033	-0.108	0.019	0.172
10m	Biodiversity x elevation: Abundance-weighted Faith's PD	env_elev	-0.148	0.066	-0.277	-0.019	0.025
10m	Biodiversity x elevation: Abundance-weighted Faith's PD	env_tri11	-0.003	0.025	-0.052	0.046	0.903
10m	Biodiversity x elevation: Abundance-weighted Faith's PD	phy_afaith	0.416	0.114	0.193	0.639	<0.001
10m	Biodiversity x elevation: Abundance-weighted Faith's PD	phy_afaith:env_elev	0.145	0.168	-0.185	0.474	0.390
10m	Biodiversity x elevation: Faith's PD	env_elev	-0.171	0.039	-0.247	-0.096	<0.001
10m	Biodiversity x elevation: Faith's PD	env_tri11	-0.010	0.025	-0.059	0.039	0.689
10m	Biodiversity x elevation: Faith's PD	phy_faith	0.113	0.037	0.040	0.185	0.003
10m	Biodiversity x elevation: Faith's PD	phy_faith:env_elev	0.050	0.052	-0.052	0.152	0.336
10m	Biodiversity x elevation: Shannon diversity	env_elev	-0.185	0.032	-0.248	-0.121	<0.001
10m	Biodiversity x elevation: Shannon diversity	env_tri11	-0.008	0.025	-0.057	0.041	0.750
10m	Biodiversity x elevation: Shannon diversity	sp_shannon	-0.038	0.033	-0.103	0.027	0.253
10m	Biodiversity x elevation: Shannon diversity	sp_shannon:env_elev	0.042	0.037	-0.031	0.114	0.261

Correlation table excerpt

Scale	Response	Predictor	n	Pearson r	Spearman r	p
10m	SA entropy mean	Faith's PD	1587	0.088	0.106	<0.001
10m	SA entropy mean	Abundance-weighted Faith's PD	1587	0.102	0.080	<0.001
10m	SA entropy mean	Shannon diversity	1587	-0.051	-0.019	0.043
10m	SA entropy mean	Elevation	1587	-0.177	-0.189	<0.001
10m	SA entropy mean	TRI 11x11	1587	-0.046	-0.051	0.070
10m	PCA mean distance	Faith's PD	1440	0.023	0.019	0.389
10m	PCA mean distance	Abundance-weighted Faith's PD	1440	0.043	0.024	0.099
10m	PCA mean distance	Shannon diversity	1440	-0.038	-0.048	0.152
10m	PCA mean distance	Elevation	1440	-0.025	-0.069	0.344
10m	PCA mean distance	TRI 11x11	1440	-0.076	-0.052	0.004
10m	Spectral Rao's Q	Faith's PD	1440	-0.008	0.002	0.760
10m	Spectral Rao's Q	Abundance-weighted Faith's PD	1440	0.003	0.007	0.907
10m	Spectral Rao's Q	Shannon diversity	1440	-0.059	-0.057	0.025
10m	Spectral Rao's Q	Elevation	1440	-0.045	-0.046	0.085
10m	Spectral Rao's Q	TRI 11x11	1440	-0.056	-0.039	0.035
10m	Alpha-hull area	Faith's PD	1439	0.023	0.001	0.377
10m	Alpha-hull area	Abundance-weighted Faith's PD	1439	0.048	0.023	0.071
10m	Alpha-hull area	Shannon diversity	1439	-0.043	-0.050	0.100
10m	Alpha-hull area	Elevation	1439	-0.003	-0.037	0.918
10m	Alpha-hull area	TRI 11x11	1439	-0.065	-0.026	0.014
20m	SA entropy mean	Faith's PD	405	0.140	0.211	0.005
20m	SA entropy mean	Abundance-weighted Faith's PD	405	0.166	0.124	<0.001
20m	SA entropy mean	Shannon diversity	405	-0.065	-0.020	0.195
20m	SA entropy mean	Elevation	405	-0.294	-0.302	<0.001